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Institution: Federal University of Petroleum Resources, Effurun
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Levels: 400

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Paper 1 of 2 — CHM 432 Petrochemistry 2024/2025

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA STATE
COLLEGE OF SCIENCE
DEPARTMENT OF CHEMISTRY
FIRST SEMESTER EXAMINATION, 2024/2025 SESSION
COURSE TITLE: PETROCHEMISTRY COURSE CODE: CHM432 UNIT: 2
INSTRUCTION ANSWER THREE QUESTIONS ONLY APRIL 2025
TIME: 2HOURS

QUESTION ONE

- a. What is steam reforming? (3 marks)
- b. Explain four impacts of sulphur and the need for desulphurization in steam reforming (16 marks)
- c. Briefly explain the importance of steam reforming (6 marks).

QUESTION TWO

- a(i) What is synthetic rubber (2 marks) (ii) Differentiate between nitrile rubber and silicone rubber and give three applications of any of them (7 marks)
- b(i) State four rules that classifies a hydrocarbon as an aromatic compound (4 marks)
- ii) State one use of each of the following aromatic hydrocarbons: methylbenzene, naphthalene, phenanthrene, trinitrotoluene (4 marks)
- (c) In a tabular form, indicate three (3) methods by which C4 streams in petrochemicals could be separated and state the function of each method (8 marks).

QUESTION THREE

- (a) List three C₅-C₈ each of unsaturated and aromatic hydrocarbon intermediates used in the petrochemical industry. 6marks
- (b) What are the sources of the following petroleum intermediates: i. Methane ii. Ethane iii. Ethylene iv. Propane v. Propylene vi. Cyclohexane 6marks
- (c) With the aid of a schematic diagram describe the production process of butadiene from crude oil. 13marks

QUESTION FOUR

- (a) State the four mechanisms involved in the transformation of organic molecules used in synthesis to create new complex molecules with applications in plastics and other products. 4marks
- (b) Explain the mechanism for the substitution of a benzene ring during synthesis. 9marks
- (c) With the aid of chemical equations or diagram explain the transformation of benzene to benzene sulphonic acid 12marks

QUESTION FIVE

- (a) Why is the acid anhydride, (RCO)₂O preferred to the acyl chloride, RCOCl in the condensation reaction with benzene to form the aromatic ketone? 3marks
- (b) With the aid of suitable reaction mechanisms and steps, explain the polymerization of ethylene to polyethylene. 12marks
- (c) State a single step process for the production of five named products from ethylene. 10marks

END OF THE PAPER

Paper 2 of 2 — PETROCHEMISTRY CHM432

Level 400 • First Semester • Handout

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
DELTA STATE, NIGERIA

ECONOMICS OF THE ORGANIC CHEMICAL INDUSTRY (CHM 432)

BY

INDUSTRIAL CHEMISTRY STUDENT
(GROUP 4)

PRESENTED TO

MR. SAMUEL ANWAMI

MARCH 2025

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ABSTRACT

This report analyzes the economic landscape of the organic chemical industry, examining key market drivers, production costs, and pricing trends. It investigates the impact of evolving consumer demand, regulatory changes, and technological advancements on the industry's growth and profitability. The analysis includes an assessment of supply chain dynamics, competitive pressures, and the potential for sustainable development within the sector.

It explores the economic implications of the growing shift towards sustainable practices within

the organic chemical industry. It investigates the financial viability of bio-based feedstocks, green chemistry processes, and circular economy models. The report also examines the role of innovation and research and development in driving cost reduction and enhancing the competitiveness of organic chemical products in a rapidly evolving market.

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INDUSTRIAL OVERVIEW OF ORGANIC CHEMICAL INDUSTRY.

The organic chemical industry is a vital sector within the global economy, responsible for producing a wide range of chemicals derived from carbon-based sources. These chemicals are fundamental to various industries, including pharmaceuticals, agriculture, textiles, and consumer goods. The industry is primarily driven by advancements in chemical engineering, biotechnology, and sustainability initiatives, shaping its economic landscape.

The organic chemical industry can be broadly categorized into:

1. Petrochemicals: Derived from crude oil and natural gas, including ethylene, propylene, and benzene, which serve as feed-stocks for plastics, synthetic rubber, and fibers.
2. Fine Chemicals: Specialty chemicals with high purity used in pharmaceuticals, Agrochemical, and electronics.
3. Bio-based Chemicals: Sourced from renewable materials such as biomass and plant-based feed stocks, supporting sustainable development.

The industry is highly globalized, with major production hubs in North America, Europe, and Asia-Pacific. Key players include multinational corporations like BASF, Dow Chemical, and Sinopec, alongside numerous smaller firms specializing in niche markets. There also production industries in Nigeria which includes Indorama Eleme Petrochemicals Limited, NNPC polypropylene and carbon black plant, and the NNPC petrochemical linear alkyd-benzene plant, Rivers, Delta and Kaduna states respectively.

FUNDMENTALS OF THE ORGANIC CHEMISTRY INDUSTRY

The organic chemical industry is a dynamic and evolving sector, with economic opportunities shaped by resource availability, innovation, and regulatory trends. The future of the industry lies in sustainable solutions, circular economy initiatives, and the development of bio-based alternatives to reduce reliance on fossil fuels.

Several key fundamentals shape the economics and operations of the organic chemical industry.

1. Raw Material Dependence: The industry heavily relies on fossil fuels and bio-based raw materials. Fluctuations in oil and natural gas prices directly impact production costs and profitability.
2. Economies of Scale: Large-scale production significantly reduces costs, leading to market dominance by a few major firms. Investments in infrastructure, technology, and automation further enhance efficiency.
3. Supply Chain and Logistics: Efficient transportation and storage of chemicals are critical due to their hazardous nature. Global supply chains play a key role in maintaining stable production and distribution.

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4. Regulatory Compliance: Stringent environmental and safety regulations, such as REACH (Registration, Evaluation, Authorisation and Restriction of Chemicals) (EU) and EPA (Environmental Protection Agency) (USA), impact production processes, requiring companies to invest in sustainable practices and waste management.

5. Technological Advancements Innovations in catalysis, process engineering, and green chemistry are driving improvements in efficiency, reducing environmental impact, and creating new market opportunities.

6. Market Demand and Growth: The industry experiences consistent growth due to rising demand in sectors like healthcare, construction, and consumer goods. However, market dynamics are influenced by economic cycles, trade policies, and global events.

MANUFACTURING PROCESS.

The manufacturing process in an organic chemical industry typically follows these key stages:

7. Raw Material Sourcing – Organic chemicals are derived from petroleum, natural gas, coal, biomass, or other sources. Common raw materials include hydrocarbons (e.g., ethylene, benzene), alcohols, and organic acids.

8. Reaction/Conversion – Chemical reactions are carried out using catalysts, heat, or pressure to transform raw materials into desired products. Examples include polymerization, oxidation, esterification, alkylation, and hydrolysis.

9. Separation & Purification – The reaction mixture often contains unreacted raw materials, by-products, and impurities. Techniques like distillation, crystallization, filtration, or solvent extraction are used for purification.

10. Formulation & Blending – Some organic chemicals, like pharmaceuticals, dyes, or agrochemical, require blending with additives to enhance performance or stability.

11. Packaging & Storage – The final product is packed in suitable containers (drums, bags, or bottles) and stored under controlled conditions before distribution.

12. Quality Control & Safety – Testing ensures the product meets industry standards, while waste treatment and emission control ensure environmental compliance.

It is important to know that these steps varies in the different organic chemical industries like petrochemicals, pharmaceuticals, or agrochemical industries

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RAW MATERIALS AND FEEDSTOCK IN THE ORGANIC CHEMICAL INDUSTRY

The organic chemical industry is a backbone of modern industrial economies, providing essential raw materials for various sectors, including pharmaceuticals, plastics, textiles, and agrochemical. The availability and processing of feed-stock directly impact production costs,

environmental sustainability, and global trade patterns.

The industry relies primarily on three broad categories of feed-stocks:

- I. Fossil Fuel-Based Feed-stock – Derived from crude oil, natural gas, and coal.
- II. Biomass-Based Feed-stock – Sourced from plant-based materials, including agricultural waste and forestry products.
- III. Recycled and Waste-Based Feed-stock – Includes industrial by-products and post-consumer recycled materials.

CLASSIFICATION OF RAW MATERIALS AND FEEDSTOCK

Raw materials for the organic chemical industry can be categorized based on their source, sustainability, and application. A comparison of key features is provided below:

Categories	Examples	Advantages	Challenges
Fossil Fuel-Based	Crude oil, natural gas, coal	High yield, well-developed infrastructure	environmental concerns
Biomass-Based	Starch, lignin, vegetable oils	Renewable, biodegradable	Seasonal availability, conversion efficiency
Recycled-Base	Plastic waste, industrial residues	Waste reduction, energy-efficient	Variable quality, collection complexity

These categories serve as the foundation for industrial chemical production.

CLASSIFICATION OF FEED-STOCK

1. Fossil fuel-based feed-stock

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Crude Oil and Petroleum Derivatives: Crude oil is the dominant feed-stock in the organic chemical industry. It is refined through fractional distillation, yielding essential raw materials such as;

Naphtha: Used in ethylene and propylene production.

Aromatics (Benzene, Toluene, Xylene - BTX): Used in resins, plastics, and synthetic fibres.

Processing Techniques:

Steam Cracking: Converts hydrocarbons into olefins.

Catalytic Reforming: Enhances octane rating and aromatics production.

Natural Gas and Shale Gas

Natural gas, primarily methane, is a key feed-stock for:

Methanol production: Used in acetic acid and formaldehyde manufacturing.

Hydrogen and ammonia production: Critical for fertilizers and synthetic fuel synthesis.

Technological Advancements:

Shale gas revolution in the US has boosted ethane-based ethylene production.

Coal-Based Chemicals

Coal is converted into useful chemicals through gasification and pyrolysis. Major products include:

Syngas ($\text{CO} + \text{H}_2$): Precursor for methanol and Fischer-Tropsch liquids.

Coal tar derivatives: Source of phenols, benzene, and naphthalene.

2. Biomass-Based Feed-stock

Agricultural and Forestry Resources

Renewable feedstocks include:

Starch (Corn, Sugarcane): Fermented to bioethanol.

Cellulose & Lignin (Wood, Straw): Used in bioplastics and biofuels.

Conversion methods include:

Biorefinery techniques: Convert agricultural waste into platform chemicals.

Enzymatic hydrolysis: Produces fermentable sugars for biochemicals.

Algae and Microbial Feedstocks

Algae and microbes offer sustainable chemical production by:

Producing biofuels (lipids to biodiesel).

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Generating biopolymers (polyhydroxyalkanoates – PHA).

Advancements in metabolic engineering allow microbes to convert CO_2 into valuable chemicals.

3. Recycled and Waste-Based Feed-stock

Plastic Waste and Industrial By-Products

Chemical recycling methods:

Pyrolysis: Converts plastics into hydrocarbon fuels.

Depolymerization: Breaks PET into reusable monomers.

Circular Economy and Sustainable Feedstock Initiatives

Industries are shifting toward waste vaporization * Catalytic Reforming: Enhances octane rating and aromatics production.

Natural Gas and Shale Gas

Natural gas, primarily methane, is a key feedstock for:

Methanol production: Used in acetic acid and formaldehyde manufacturing.

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Coal tar derivatives: Source of phenols, benzene, and naphthalene.

PRODUCTION PROCESS

The production process in an organic chemical industry involves a series of steps to transform

raw materials into finished products. The key stages include:

1. Feed-stock Preparation – Raw materials (such as petroleum fractions, natural gas, or biomass) are pre-treated to remove impurities before entering the reaction stage.
2. Chemical Reaction – The core transformation occurs through processes like polymerization, esterification, oxidation, hydrogenation, or alkylation, depending on the desired product. Catalysts, temperature, and pressure are carefully controlled.

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3. Separation & Purification – The reaction mixture contains the main product, unreacted materials, and by-products. Distillation, crystallization, filtration, or extraction help isolate the pure product.

4. Product Refinement – Some products require further processing, such as drying, neutralization, or blending with additives, to meet required specifications.

5. Quality Control – Analytical techniques like spectroscopy, chromatography, or titration ensure the product meets industry standards.

6. Packaging & Distribution – The final product is packaged in appropriate containers and stored under suitable conditions before being distributed to customers.

VOLATILITY IN RAW MATERIAL PRICES

Chemical production requires various raw materials, including oil, natural gas, and minerals. The prices of these materials can be volatile, making it difficult for companies to predict costs and manage supply chain risks.

GROWING DEMANDS FOR CHEMICALS

The demand for chemicals will continue to rise as the global population grows and urbanizes. Chemicals are used in almost every aspect of modern life, from construction to transportation to healthcare, making this industry an essential part of the global economy.

EFFECTS OF CHANGING ENERGY.

Energy sources can have significant effects on the environment, that economy, and society, primarily impacting climate change through reduced greenhouse gas emissions when transitioning to renewable sources but also potentially causing disruptions in industries reliant on fossil fuels, job shifts, and changes to infrastructure depending on the new energy sources adopted; overall, a shift to cleaner energy aims to mitigate environmental damage while potentially creating new economic opportunities.

REGULATIONS AND POLICIES RELATED TO EMISSIONS AND WASTE MANAGEMENT

International Regulations and Policies

1. Basel Convention (1989): Regulates transboundary movement and disposal of hazardous waste.
2. Stockholm Convention (2001): Regulates persistent organic pollutants (POPs) emissions.
3. Kyoto Protocol (1997): Sets greenhouse gas emissions reduction targets.

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4. Paris Agreement (2015): Aims to limit global warming to well below 2°C.

National Regulations and Policies

1. United States

- Clean Air Act (1970): Regulates emissions from stationary and mobile sources.
- Resource Conservation and Recovery Act (RCRA) (1976): Manages hazardous waste.
- Toxic Substances Control Act (TSCA) (1976): Regulates toxic substances.

2. European Union

- Waste Framework Directive (2008/98/EC): Sets waste management targets and guidelines.
- Emissions Trading System (EU ETS): Caps greenhouse gas emissions from power and industry sectors.

3. China

- Air Pollution Prevention and Control Law (2015): Regulates air pollutant emissions.
- Solid Waste Environmental Pollution Prevention Law (2019): Manages solid waste.

Emission-Specific Regulations

1. Carbon Emissions

- Carbon Pricing: Mechanisms like carbon taxes or cap-and-trade systems.
- Renewable Portfolio Standards (RPS): Require utilities to generate a percentage of electricity from renewable sources.

2. Air Pollutants

- National Ambient Air Quality Standards (NAAQS): Set limits for criteria pollutants like particulate matter, ozone, and nitrogen dioxide.
- Emissions Standards: Regulate emissions from industrial sources, vehicles, and other sources.

Waste Management-Specific Regulations

1. Waste Disposal

- Landfill Regulations: Govern landfill design, operation, and closure.
- Incineration Regulations: Regulate waste combustion and emissions.

2. Waste Recycling

- Recycling Targets: Set goals for recycling rates.
- Extended Producer Responsibility (EPR): Require manufacturers to take responsibility for waste generated by their products.

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Policy Instruments

1. Command-and-Control Regulations: Directly regulate emissions and waste management practices.
2. Market-Based Instruments: Use economic incentives, like taxes or tradable permits, to encourage sustainable practices.
3. Voluntary Agreements: Encourage industries to adopt sustainable practices through voluntary commitments.

Emerging Trends and Technologies

1. Circular Economy: Encourages waste reduction, reuse, and recycling.

2. Carbon Capture and Utilization: Develops technologies to capture and convert CO₂ into valuable products.
3. Digitization and IoT: Leverages digital technologies to optimize waste management and emissions reduction.

TECHNOLOGICAL ADVANCEMENT

The chemical industry is constantly evolving, and new technologies are being developed to increase efficiency, reduce waste, and improve safety. Innovations such as green chemistry, process intensification, and 3D printing are changing how chemicals are produced and consumed.

OPPORTUNITIES.

The organic chemical industry offers diverse opportunities across various sectors. Some key areas include:

1. Manufacturing & Production
 - i. Petrochemicals (plastics, synthetic fibers, rubber)
 - ii. Pharmaceuticals (drug synthesis, intermediates)
 - iii. Agrochemicals (fertilizers, pesticides, herbicides)
 - iv. Food & Beverages (flavors, preservatives, additives)
 - v. Cosmetics & Personal Care (fragrances, skincare ingredients)
2. Research & Development (R&D)
 - i. New material synthesis (biodegradable plastics, green chemicals)
 - ii. Process optimization (catalysts, energy-efficient reactions)
 - iii. Drug discovery & formulation in pharmaceuticals
 - iv. Sustainable & bio-based chemical development
3. Quality Control & Regulatory Compliance
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 - i. Testing & analysis (GC-MS, HPLC, spectroscopy)
 - ii. Ensuring compliance with industry standards (ISO, FDA, REACH)
 - iii. Environmental monitoring & waste management
4. Sales & Marketing
 - i. Chemical product distribution
 - ii. Business development for industrial clients
 - iii. Market analysis for emerging chemical trends
5. Sustainability & Green Chemistry
 - i. Biofuels & renewable chemicals
 - ii. Waste recycling & pollution control
 - iii. Carbon capture & utilization technologies
6. Academia & Consulting
 - i. Teaching & training future chemists
 - ii. Industrial consulting on process improvement & innovation
 - iii. Policy development for chemical safety & regulations

CHALLENGES.

The chemical industry plays a crucial role in virtually every aspect of modern life, from the clothes we wear to the food we eat to the devices we use to communicate. The chemical industry faces numerous challenges in meeting the demands of a rapidly changing world while ensuring the safety of the environment and the well-being of its workers

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CONCLUSION

The organic chemical industry is undergoing a transition from fossil-based feedstock to more sustainable alternatives. Advances in green chemistry, circular economy models, and biotechnological innovations will shape the future of feedstock utilization.

The chemical industry is a dynamic and growing sector essential to the global economy. While it faces significant challenges, suitable investments and strategies can help overcome them. The industry is growing, and businesses have many opportunities to grow and expand. Safety, sustainability, and innovation are key factors in the industry's success, and companies must continue to invest in these areas to remain competitive. By embracing these challenges and opportunities, the chemical industry can continue to contribute to the global economy and positively impact the world.

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- Kirk-Othmer Encyclopedia of Chemical Technology (Wiley)
- Market research reports from Statista, Grand View Research, or McKinsey & Company

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES,
EFFURUN, DELTA STATE NIGERIA.

C4 (CARBON 4) STREAMS FROM NAPHTHA CRACKERS

BY

DEPARTMENT OF CHEMISTRY

MARCH 2025

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ABSTRACT

A method for separation of a crude C₄ cut, containing butane, butene, 1,3-butadiene and small amounts of other hydrocarbons, amongst which are C₂, acetylenes, by means of extractive distillation with a selective solvent is disclosed, whereby the crude C₄ cut is introduced into a first extractive distillation column (KI) in the mid region thereof, the selective solvent introduced above the crude C₄ cut and a vapour side stream withdrawn from the first extractive distillation column below the introduction point of the crude C₄ cut, containing the C₂, acetylenes and predominantly the selective solvent as well, whereby the concentration of the C₂, acetylenes in the vapour side stream lies below the spontaneous decomposition level and a head stream is withdrawn containing the components which are less soluble than the C₂, acetylenes in the selective solvent.

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DEFINITION

C₄ streams refer to the fraction of hydrocarbons containing four carbon atoms (C₄) that are produced as byproducts in naphtha cracking during the production of ethylene and propylene in steam crackers. These streams contain a mix of saturated and unsaturated C₄ hydrocarbons, which serve as valuable feedstocks for petrochemical and refining processes.

COMPOSITION

The composition of C₄ streams varies depending on the feedstock and operating conditions of the naphtha cracker but typically includes:

1. Alkanes (Saturated C₄s):

o n-Butane (C₄H₁₀) – Used as an LPG component or feedstock for isomerization.

o Isobutane (C₄H₁₀) – Used in alkylation units for gasoline production and as a feedstock for isobutylene production.

2. Alkenes (Unsaturated C₄s):

- o 1-Butene (C₄H₈) – Used in polyethylene production, synthetic lubricants, and plasticizers.
- o 2-Butene (C₄H₈) (cis and trans isomers) – Used in chemical synthesis and polymerization.
- o Isobutylene (C₄H₈) – Used in the production of methyl tert-butyl ether (MTBE), butyl rubber, and polyisobutylene.

3. Dienes (Highly Reactive C₄s):

- o 1,3-Butadiene (C₄H₆) – A key raw material for synthetic rubber, latex, and ABS plastics.
- o Other C₄ dienes (small quantities) – Potential impurities or intermediates in chemical processing.

SOURCE

C₄ streams originate from naphtha crackers, which thermally break down naphtha into lighter hydrocarbons using steam cracking. This process generates various byproducts, including hydrogen, methane, ethylene, propylene, C₄ hydrocarbons, and pyrolysis gasoline. The C₄ fraction is separated and processed further for industrial use.

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KEY COMPONENTS

The typical composition of C₄ streams includes:

- Butanes (n-butane, isobutane) – Saturated alkanes used in LPG, alkylation, and isomerization.
- Butenes (1-butene, 2-butene, isobutene) – Unsaturated alkenes used in polymer production and fuel blending.
- Butadiene (1,3-butadiene) – A highly valuable conjugated diene used in synthetic rubber manufacturing.
- Methylacetylene and Propadiene (MAPD) – Minor constituents that require removal for downstream processing.

INDUSTRIAL APPLICATIONS

C₄ hydrocarbons play a significant role in multiple industries:

1. Butadiene (C₄H₆) Applications

- Synthetic Rubber Production – Used in styrene-butadiene rubber (SBR) and polybutadiene rubber (PBR) for tires, footwear, and industrial belts.
- Plastics & Resins – Key raw material for Acrylonitrile Butadiene Styrene (ABS) and Styrene-Butadiene Latex (SBL).
- Nylon Production – Intermediate in adiponitrile synthesis for nylon-6,6 fibers.

2. Butene's (C₄H₈) Applications

- 1-Butene – Used in polyethylene copolymers for flexible packaging and pipes.
 - 2-Butene – Utilized in chemical synthesis and refining processes.
 - Isobutene (Isobutylene) –
- o Precursor for methyl tert-butyl ether (MTBE) and tert-amyl methyl ether

(TAME) (gasoline octane boosters).

- o Produces butyl rubber, essential for tire inner linings and industrial gaskets.

3. Butanes (C₄H₁₀) Applications

- n-Butane – Used as liquefied petroleum gas (LPG), as a feedstock for maleic anhydride, and in petrochemical cracking.

- Isobutane –

- o Used in alkylation units to produce high-octane gasoline.

- o Acts as a refrigerant (R-600a) in eco-friendly cooling systems.

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4. Other Industrial Uses

- Petrochemical Feedstocks – C₄ fractions are key intermediates in producing higher hydrocarbons, solvents, and lubricants.

- Fuel Additives – MTBE, TAME, and iso-octane derivatives enhance gasoline performance.

- Aerosols & Propellants – Butane and isobutane are used in aerosol sprays and foam blowing agents.

SEPARATION AND FRACTIONATION

C₄ streams from naphtha crackers contain a mixture of butadiene, butenes, butanes, and other minor components. Efficient separation is necessary to isolate high-purity fractions for industrial applications. The key separation processes include:

1. Initial C₄ Stream Separation

After cracking, C₄ hydrocarbons are separated from lighter gases (C₁–C₃) and heavier fractions (C₅+). The primary separation methods include:

- Compression & Cooling – Condenses C₄s into a liquid phase.

- Primary Fractionation (Depropanizer & Debutanizer) –

- o Depropanizer removes lighter gases (C₃–).

- o Debutanizer separates C₄s from heavier fractions (C₅+).

2. Butadiene Extraction

Since butadiene is a highly valuable component, it is typically extracted first. The main processes include:

a) Extractive Distillation (Solvent-Based)

- Uses selective solvents such as N-Methyl-2-pyrrolidone (NMP), dimethylformamide (DMF), or furfural to separate butadiene from butene's and butanes.

- Solvent absorbs butadiene, and steam stripping or vacuum distillation recovers pure butadiene.

b) Distillative Separation (Cryogenic or Conventional Distillation)

- High-purity butadiene is recovered via super fractionation using multiple distillation columns.

- Typically, a butadiene extractive distillation column followed by a butadiene purification column ensures >99% purity.

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3. Separation of Butene's & Butanes

After butadiene extraction, the remaining C4 mixture (butene's & butanes) is further separated using:

a) Catalytic Hydrogenation (Selective Hydrogenation)

- Converts trace butadiene into butene's or butanes to stabilize the C4 stream.
- Common catalysts: Palladium-based catalysts.

b) Fractionation of Butene's (Butene-1, Butene-2, and Isobutene)

- Isobutene Separation: Extracted via distillation or molecular sieve adsorption.
- 1-Butene vs. 2-Butene: Separated by fractional distillation based on boiling point differences.
- Applications:

o 1-Butene → Polyethylene co-monomers.

o Isobutene → MTBE, butyl rubber.

c) Butane Separation (n-Butane vs. Isobutane)

- Distillation (Super fractionation): Separates n-butane (BP: -0.5°C) from isobutane (BP: -11.7°C).
 - Applications:
- o Isobutane → Alkylation for high-octane gasoline.
- o n-Butane → LPG, solvent applications.

4. Purification & Final Treatment

To achieve high-purity products, additional purification steps include:

- Caustic Washing & Drying: Removes acidic impurities (H_2S , CO_2).
- Molecular Sieves & Adsorption: Eliminates residual moisture and trace contaminants.

Summary of Key Technologies Used

Process Purpose Key Methods

Debutanization Removes C5+ from C4s Fractional distillation

Extractive Distillation Butadiene extraction NMP, DMF, or furfural solvents

Selective Hydrogenation Converts butadiene to butenes Pd-based catalysts

Superfractionation Separates butenes and butanes High-purity distillation

Adsorption (Molecular Sieves) Removes trace impurities Zeolite-based adsorption

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ECONOMIC IMPORTANCE

C4 streams are highly valuable in the petrochemical industry, serving as key feedstocks for rubber, plastics, fuel additives, and specialty chemicals. Their economic significance arises from their widespread industrial applications and their impact on global supply chains.

1. High-Value Derivatives & Market Demand

Each component of the C4 stream contributes significantly to different industries:

Component Main Product Industry Market Value

Applications

Butadiene (C₄H₆) Synthetic rubber (SBR, PBR) Tires, auto parts, High due to demand in the footwear automotive sector

1-Butene Polyethylene co-monomer Packaging, pipes Growing with plastic & packaging demand

Isobutene MTBE, butyl rubber Fuel additives, tire Driven by fuel and manufacturing automotive markets

n-Butane & LPG, Alkylation feedstock Fuel, refrigerants, Influenced by energy and Isobutane aerosols refining industries

2. Impact on the Automotive & Tire Industry

- Butadiene is crucial for synthetic rubber production, particularly SBR & PBR, which account for ~70% of the global rubber demand.
- The tire industry heavily relies on butadiene-based rubbers, making C4 streams integral to the automotive supply chain.
- Global demand for high-performance tires (e.g., low-rolling-resistance tires) increases the value of butadiene-derived rubbers.

3. Role in the Plastics & Packaging Industry

- 1-Butene is a key co-monomer for LLDPE (linear low-density polyethylene), used in flexible packaging, films, and containers.
- Growth in e-commerce, food packaging, and consumer goods drives demand for butene-based polyethylene.

4. Influence on the Energy & Fuel Sector

- Isobutene is used in MTBE (Methyl Tert-Butyl Ether), a gasoline additive that enhances octane ratings.

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- Butanes (n-butane & isobutane) are essential in LPG (liquefied petroleum gas), widely used as fuel in residential and industrial applications.
- Isobutane plays a role in alkylation units, where it is converted into high-octane gasoline components.

5. Market Trends & Price Sensitivity

- C4 stream prices are highly volatile, driven by crude oil and naphtha prices, as well as demand fluctuations in end-use industries.
- Butadiene prices are cyclical, influenced by supply-demand imbalances, cracker turnarounds, and automotive sales trends.
- Shale gas developments (U.S.) have impacted global C4 supply:
 - o Ethane-based crackers produce fewer C4s, tightening supply.
 - o This increases reliance on naphtha cracking in Asia & Europe for C4 feedstocks.

6. Regional Market Dynamics

- Asia-Pacific dominates the C4 market, with China leading in synthetic rubber production.
- North America benefits from shale gas, but relies on imports for butadiene due to lighter feedstock cracking.

- Europe & Middle East maintain strong refining & petrochemical integration, securing a steady C4 supply.

ENVIRONMENTAL CONSIDERATIONS

The processing and utilization of C4 hydrocarbons have several environmental impacts. Key concerns include air pollution, greenhouse gas emissions, hazardous waste generation, and energy consumption. However, advancements in process optimization and sustainability initiatives are helping to mitigate these effects.

1. Air Pollution & Emissions

C4 processing and downstream applications contribute to volatile organic compound (VOC) emissions, greenhouse gases (GHGs), and toxic pollutants.

a) VOC & Hydrocarbon Emissions

- Butadiene & Butenes are reactive VOCs (Ozone Precursors)
 - o Can contribute to ground-level ozone formation and smog.
 - o Classified as hazardous air pollutants (HAPs) by the U.S. EPA.

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- o Strict emission controls (e.g., thermal oxidizers, vapor recovery units) are required in industrial plants.

- Butadiene is a known carcinogen

- o Long-term exposure linked to leukemia and respiratory issues.
- o Regulations enforce leak detection and repair (LDAR) programs to minimize emissions.

b) Greenhouse Gas (GHG) Emissions

- High energy demand in separation & purification units leads to CO₂ emissions from:
 - o Steam cracking & distillation processes
 - o Hydrogenation & alkylation units
- Use of carbon capture, utilization, and storage (CCUS) in petrochemical plants is growing to mitigate CO₂ emissions.

2. Wastewater & Hazardous Waste Generation

- Solvent-based butadiene extraction (NMP, DMF, or furfural) generates wastewater with organic contaminants.
 - o Requires biological treatment, activated carbon filtration, or advanced oxidation for safe discharge.
- Spent catalysts from hydrogenation & polymerization (e.g., palladium-based) require proper disposal or recycling.

3. Energy Consumption & Process Efficiency

- C4 separation (extractive distillation, cryogenic fractionation) is energy-intensive.
 - o Adoption of heat integration & process optimization helps reduce energy consumption.
- Membrane separation & adsorption technologies are emerging as lower-energy alternatives.

4. Safer Alternatives & Green Chemistry Approaches

- MTBE (from isobutene) is being phased out in some regions due to groundwater contamination risks.
- o Replacements like ETBE (ethyl tert-butyl ether) and bio-based fuel additives are gaining popularity.
- Development of bio-based butadiene & butenes from renewable sources (e.g., biomass, fermentation) could reduce reliance on fossil feedstocks.

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5. Regulatory & Sustainability Efforts

- Stringent Environmental Regulations:
 - o U.S. EPA Clean Air Act and European REACH regulations limit butadiene and VOC emissions.
 - o Emission trading systems (ETS) and carbon taxes are pushing the industry toward greener technologies.
- Circular Economy & Recycling:
 - o Chemical recycling of butadiene-based plastics & synthetic rubber is being explored.
 - o End-of-life tire recycling & pyrolysis could recover valuable C4 hydrocarbons.

MARKET DEMAND

The global market demand for C4 derivatives from naphtha crackers, particularly butadiene, isobutene, and 1-butene, is experiencing significant growth, driven by their extensive industrial applications.

1. Butadiene Market

- Market Size & Growth: The global butadiene market was valued at approximately USD 12.48 billion in 2024 and is projected to reach around USD 30.37 billion by 2034, expanding at a CAGR of 9.30% during this period.
- Key Drivers:
 - o Automotive Industry: High demand for synthetic rubbers, such as Styrene-Butadiene Rubber (SBR) and Polybutadiene Rubber (PBR), used predominantly in tire manufacturing.
 - o Construction Sector: Utilization of butadiene-based polymers in various applications.

2. Isobutene Market

- Market Size & Growth: The global isobutene market was valued at USD 32.45 billion in 2025 and is projected to reach USD 49.48 billion by 2033, growing at a CAGR of 4.8% during the forecast period.
- Key Drivers:
 - o Fuel Additives: Production of Methyl Tert-Butyl Ether (MTBE), enhancing gasoline octane ratings.

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o Synthetic Rubber: Manufacture of butyl rubber, essential for tire inner linings and various industrial applications.

3. 1-Butene Market

- Market Size & Growth: The 1-butene market was valued at USD 806.83 million in 2023, expected to reach USD 854.63 million in 2024, and is projected to grow at a CAGR of 5.11%, reaching USD 1,143.65 million by 2030.

- Key Drivers:

- o Polyethylene Production: Use as a co-monomer in producing Linear Low-Density Polyethylene (LLDPE), widely used in flexible packaging solutions.

- o Adhesives & Sealants: Incorporation into various formulations for enhanced performance.

Regional Insights

- Asia-Pacific: Dominates the butadiene market, with significant growth in automotive manufacturing and construction activities.

- North America & Europe: Steady demand due to established automotive industries and ongoing infrastructural projects.

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CONCLUSION

C4 streams from naphtha crackers are vital to the petrochemical industry, serving as key feedstocks for synthetic rubber, plastics, fuel additives, and specialty chemicals. Their economic importance is driven by demand in the automotive, packaging, and energy sectors, with strong market growth projected for butadiene, isobutene, and 1-butene.

However, environmental concerns such as VOC emissions, hazardous waste, and high energy consumption pose challenges. Regulatory frameworks and sustainability initiatives including process optimization, carbon capture, and bio-based alternatives are shaping a more sustainable future for C4 processing.

Market dynamics are influenced by regional supply chains, crude oil prices, and shifting feedstock trends (e.g., shale gas vs. naphtha cracking). The Asia-Pacific region leads demand, while North America and Europe focus on technological advancements and regulatory compliance.

Overall, C4 streams will continue to play a crucial role in global industrial growth, with opportunities for innovation in green chemistry, circular economy practices, and advanced separation technologies.

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S/N NAMES MATRIC NUMBER SIGN

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3 EDUMU EBIAREDE MARION COS/7371/2020

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7 ESHIEMOKHAI MARYLYN COS/7377/2020

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11 NWAIWU CHIMAOBIM NNAMDI COS/7382/2020

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STEAM REFORMING AND ITS CONSEQUENCES
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GROUP B
FEBRUARY, 2025

ABSTRACT

Steam reforming is a critical industrial process for generating hydrogen, essential for various applications including ammonia synthesis, refining, and fuel cells. This abstract provides an overview of the steam reforming process, encompassing feedstock preparation, catalytic reactions, water-gas shift, and purification stages. It also addresses the challenges associated with conventional steam reforming, particularly CO₂ emissions, and highlights advancements in technologies aimed at enhancing efficiency and sustainability.

INTRODUCTION

Steam reforming or steam methane reforming (SMR) is a method for producing syngas (hydrogen and carbon monoxide) by reaction of hydrocarbons with water. Commonly, natural gas is the feedstock. The

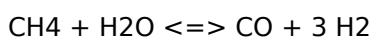
main purpose of this technology is often hydrogen production, although syngas has multiple other uses

such as production of ammonia or methanol. The reaction is represented by this equilibrium:

Illustrating inputs and

outputs of steam reforming of natural gas, a process to produce hydrogen and CO₂ greenhouse gas that

may be captured with CCS.



The reaction is strongly endothermic ($\Delta H_{\text{SR}} = 206 \text{ kJ/mol}$).

Hydrogen produced by steam reforming is termed as 'grey' hydrogen when the waste carbon dioxide is

released to the atmosphere and 'blue' hydrogen when the carbon dioxide is (mostly) captured and stored geologically—see carbon capture and storage. Zero carbon 'green' hydrogen is produced by thermochemical water splitting, using solar thermal, low- or zero-carbon electricity or waste heat, or

electrolysis, using low- or zero-carbon electricity. Zero carbon emissions 'turquoise' hydrogen is produced by one-step methane pyrolysis of natural gas.

Steam reforming of the natural gas produces most of the world's hydrogen. Hydrogen is used in the industrial synthesis of ammonia and other chemicals

IMPORTANCE:

Steam reforming is applied in large-scale industrial contexts to produce hydrogen - for example, for use

in petroleum refineries. Steam reforming is the most commonly used process for hydrogen production.

Fossil fuels such as natural gas and petroleum or coal are usually used as feedstocks. Under pressure and

high temperatures, the hydrocarbons contained in the energy sources are then converted into methane,

carbon monoxide and carbon dioxide. These substances are then catalyzed to form hydrogen.

As an alternative to fossil fuels, biomass can also be used for steam reforming. This improves the overall

CO₂ balance of the process. Because almost any type of biomass is suitable for the process, the hydrogen output is very high. The hydrogen produced by biogas steam reforming way is also referred to

as biohydrogen.

VARIOUS FEEDSTOCKS USE:

Natural gas steam reforming:

Natural gas steam reforming is a mature technology that currently generates 48% of the world's hydrogen production (Ferreira-Aparicio, Benito, & Sanz, 2005).

After treatment of natural gas from wells, it contains mainly one chemical compound, methane. The typical composition of natural gas is as follows (expressed as volume percentage): CH₄: 95%, C₂+: 3.5%,

N₂: 1%, CO₂: 0.5%, and small amounts of sulfur compounds.

In steam reforming, steam reacts with the feedstock in a set of reactions to produce mainly hydrogen, carbon dioxide, and carbon monoxide. Natural gas is the most important feedstock for steam reforming.

Main reactions of natural gas steam reforming are:

C_nH_{2n+2} in reaction (4.2) represents hydrocarbons higher than methane contained in natural gas. Reaction (4.3) is called a water gas shift (WGS) reaction. Reactions (4.1) and (4.2) are endothermic and reaction (4.3) is slightly exothermic. Globally, the process is endothermic.

Naphtha steam reforming:

The most widespread process for rearranging hydrocarbon molecules is naphtha reforming. The initial

process, thermal reforming, was developed in the late 1920s. Thermal reforming employed temperatures of 510–565 °C (950–1,050 °F) at moderate pressures—about 40 bars (4 MPa), or 600 psi—

to obtain gasolines (petrols) with octane numbers of 70 to 80 from heavy naphthas with octane numbers of less than 40. The product yield, although of a higher octane level, included olefins, diolefins,

and aromatic compounds. It was therefore inherently unstable in storage and tended to form heavy polymers and gums, which caused combustion problems.

By 1950 a reforming process was introduced that employed a catalyst to improve the yield of the most

desirable gasoline components while minimizing the formation of unwanted heavy products and coke.

(A catalyst is a substance that promotes a chemical reaction but does not take part in it.) In catalytic

reforming, as in thermal reforming, a naphtha-type material serves as the feedstock, but the reactions

are carried out in the presence of hydrogen, which inhibits the formation of unstable unsaturated compounds that polymerize into higher-boiling materials.

Importance of Feedstock Purification (Desulfurization) on Steam Reforming

1. Catalyst Protection

* Catalyst Deactivation: Sulfur compounds present in the feedstock (natural gas or other hydrocarbons)

can poison the catalyst used in steam reforming, typically nickel-based. Sulfur binds strongly to the

active sites on the catalyst surface, leading to deactivation and reduced efficiency.

* **Extended Catalyst Life:** By removing sulfur compounds, the catalyst's lifespan is extended, reducing the frequency of catalyst regeneration or replacement and thereby lowering operational costs.

2. Process Efficiency

* **Optimal Performances:** A clean feedstock ensures that the catalyst operates at optimal performance, maintaining high conversion rates of hydrocarbons to hydrogen and synthesis gas (syngas). This improves overall process efficiency and yield.

* **Preventing Side Reactions:** Sulfur and other impurities can cause unwanted side reactions that produce byproducts, reducing the purity of the hydrogen or syngas produced. Desulfurization minimizes these side reactions, leading to a purer product.

3. Environmental Benefits

* **Reduced Emissions:** Sulfur compounds in the feedstock can lead to the formation of sulfur oxides (SO_x) during combustion, contributing to air pollution and acid rain. Desulfurization helps reduce these harmful emissions, promoting a cleaner and more environmentally friendly process.

* **Compliance with Regulations:** Many regions have strict environmental regulations regarding sulfur emissions. Desulfurizing the feedstock ensures compliance with these regulations, avoiding potential fines and legal issues.

4. Safety

* **Preventing Corrosion:** Sulfur compounds can cause corrosion of equipment, leading to maintenance issues and potential safety hazards. By removing sulfur, the integrity and safety of the steam reforming infrastructure are maintained.

* **Avoiding Hazards:** Hydrogen sulfide (H₂S) is a toxic and flammable gas. Its removal from the feedstock reduces the risk of exposure to this hazardous substance, ensuring a safer working environment.

REACTIONS:

Steam reforming reaction kinetics, in particular using nickel-alumina catalysts, have been studied in detail since the 1950s.

PRE-REFORMING

Depiction of the general process

flow of a typical steam reforming plant. (PSA = Pressure swing adsorption, NG = Natural gas).

The purpose of pre-reforming is to break down higher hydrocarbons such as propane, butane or naphtha into methane (CH₄), which allows for more efficient reforming downstream.

Steam-Methane Reforming

Most hydrogen produced today in the United States is made via steam-methane reforming, a mature production process in which high-temperature steam (700°C–1,000°C) is used to produce hydrogen from

a methane source, such as natural gas. In steam-methane reforming, methane reacts with steam under

3–25 bar pressure (1 bar = 14.5 psi) in the presence of a catalyst to produce hydrogen, carbon monoxide,

and a relatively small amount of carbon dioxide. Steam reforming is endothermic—that is, heat must be

supplied to the process for the reaction to proceed.

Subsequently, in what is called the "water-gas shift reaction," the carbon monoxide and steam are reacted using a catalyst to produce carbon dioxide and more hydrogen. In a final process step called

"pressure-swing adsorption," carbon dioxide and other impurities are removed from the gas stream,

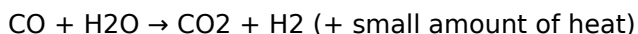
leaving essentially pure hydrogen. Steam reforming can also be used to produce hydrogen from other

fuels, such as ethanol, propane, or even gasoline.

Steam-methane reforming reaction



Water-gas shift reaction



Partial Oxidation

In partial oxidation, the methane and other hydrocarbons in natural gas react with a limited amount of

oxygen (typically from air) that is not enough to completely oxidize the hydrocarbons to carbon dioxide

and water. With less than the stoichiometric amount of oxygen available, the reaction products contain

primarily hydrogen and carbon monoxide (and nitrogen, if the reaction is carried out with air rather than

pure oxygen), and a relatively small amount of carbon dioxide and other compounds.

Subsequently, in a

water-gas shift reaction, the carbon monoxide reacts with water to form carbon dioxide and more hydrogen.

Partial oxidation is an exothermic process—it gives off heat. The process is, typically, much faster than

steam reforming and requires a smaller reactor vessel. As can be seen in chemical reactions of partial

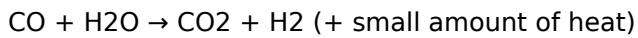
oxidation, this process initially produces less hydrogen per unit of the input fuel than is obtained by

steam reforming of the same fuel.

Partial oxidation of methane reaction

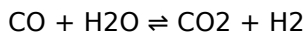


Water-gas shift reaction



THE WATER -GAS SHIFT REACTION

The water-gas shift reaction (WGSR) describes the reaction of carbon monoxide and water vapor to form carbon dioxide and hydrogen:



The water gas shift reaction was discovered by Italian physicist Felice Fontana in 1780. It was not until

much later that the industrial value of this reaction was realized. Before the early 20th century, hydrogen was obtained by reacting steam under high pressure with iron to produce iron oxide and hydrogen. With the development of industrial processes that required hydrogen, such as the Haber-

Bosch ammonia synthesis, a less expensive and more efficient method of hydrogen production was needed. As a resolution to this problem, the WGSR was combined with the gasification of coal to produce hydrogen.

GAS PURIFICATION AND SEPARATION

Pressure swing adsorption (PSA) units are used to separate impurities from hydrogen streams, producing high-purity hydrogen and a low-pressure tail gas stream containing the impurities. This process involves using beds of solid adsorbent, which are regenerated by depressurizing and purging. Up

to 20% of hydrogen may be lost in the tail gas.

PSA technology relies on the physical binding of gas molecules to adsorbent materials. The separation

effect is based on differences in binding forces, with highly volatile and low polarity components like

hydrogen being practically nonadsorbable. Impurities such as nitrogen, carbon monoxide, carbon dioxide, hydrocarbon derivatives, and water vapor are adsorbed from the hydrogen stream, leading to

the recovery of high-purity hydrogen.

The PSA process operates at a constant temperature, using alternating pressure and partial pressure for

adsorption and desorption. Adsorption occurs at high pressure (typically 10–40 bar) until equilibrium

loading is reached, after which the adsorbent material must be regenerated by lowering the pressure.

The process is cyclic, with short cycles in the range of minutes, allowing the economical removal of large amounts of impurities.

PSA is the preferred purification method for steam reforming units and is also used for refinery off-gases.

It produces higher-purity hydrogen (99.9% v/v) compared to wet scrubbing processes (<97% v/v). The

cyclic nature of PSA involves passing the gas through adsorbent beds, with minimal hydrogen adsorption,

followed by bed regeneration through depressurization and low-pressure purging.

Hydrogen purification by pressure swing adsorption

CONSEQUENCES OF STEAM REFORMING

1. Carbon Dioxide Emissions

- One of the major drawbacks of steam reforming is the significant emission of carbon dioxide (CO₂), a

greenhouse gas that contributes to global warming and climate change. The process generates CO₂ as a

byproduct, which poses environmental challenges.

2. Catalyst Deactivation

- The catalysts used in steam reforming, typically nickel-based, are prone to deactivation due to carbon

deposition (coking), sulfur poisoning, and sintering. Catalyst deactivation reduces the efficiency of the

process and requires periodic regeneration or replacement.

3. Energy Consumption

- Steam reforming is an endothermic process, requiring a substantial amount of energy to maintain the

high temperatures needed for the reaction. This energy is often derived from fossil fuels, leading to additional CO₂ emissions and energy costs

4. Operational Challenges

- The process involves high temperatures and pressures, which can lead to operational challenges such

as equipment wear and tear, tube failures, and refractory problems. These issues can result in downtime,

maintenance costs, and safety concerns.

5. Environmental Impact

- The production and transportation of natural gas, the primary feedstock for steam reforming, can have environmental impacts, including methane leaks, habitat disruption, and water contamination.

Additionally, the disposal of spent catalysts poses environmental challenges.

CONCLUSION

Steam reforming currently provides the majority of the world's industrial hydrogen, underscoring its critical role. However, the associated CO₂ emissions present a significant challenge. To maintain its

relevance in a carbon-constrained world, the industry must prioritize the integration of carbon capture technologies and the exploration of alternative, low-carbon feedstocks. By addressing these challenges, steam reforming can continue to contribute to essential industrial processes while aligning with global sustainability goals.

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PETROCHEMICALS

The petrochemicals are chemicals derived from petroleum and natural gas either by direct manufacture or indirect manufacture as byproducts from the variety of processes that are used during refining of petroleum. Gasoline, kerosene, fuel oil, lubricating oil, wax, asphalt are excluded from the definition of petrochemicals since they are not, in the true sense, chemical compounds but intimate mixtures of hydrocarbons. Some chemical compounds made from petroleum are also obtained from other fossil fuels, such as coal or natural gas, or renewable sources such as maize, palm fruit or sugar cane. So, the terminology, 'Petrochemical' is a matter of source identification.

Petrochemical industry therefore sources raw materials from refining and gas-processing and converts these raw materials into valuable products using a variety of chemical process technologies. A variety of feedstocks are used as raw materials, the industry driver being economic.

The starting materials for the Petrochemical Industry are obtained from crude oil in one of two general ways. Below is the table showing the classes of hydrocarbon intermediates used in the Petrochemical Industry.

Table 1: Hydrocarbon Intermediates used in the Petrochemical Industry

Carbon Hydrocarbon type

Number Saturated Unsaturated Aromatic

1 Methane

2 Ethane Ethylene, acetylene

3 Propane Propylene

4 n-butene, iso-butane,

Butane butadiene

5

Pentane Isopentene, isoprene

6

Hexane, cyclohexane Methyl pentene Benzene

7 Mixed heptane Toluene

8. di- iso butylene Xylene , Styrene

9 Cumene

12 Propylene tetramer

Tri- isobutylene

18 Dodecylbenzene

The above intermediates may be present in the raw crude oil as such are isolated by physical methods such as distillation and solvent extraction. On the other hand, they may be present if at all in trace amounts, and are synthesized during the refining operation. In fact the unsaturated hydrocarbons, which are not usually present in virgin (unprocessed) petroleum are nearly always manufactured during the various refining operations

1

Table 2: Sources of petroleum intermediate

Hydrocarbon Sources

Methane Natural gas

Ethane Natural gas

Ethylene Cracking process

Propane Natural gas and catalytic reforming, catalytic processes

Propylene Natural gas, reforming

Butane Cracking processes

Cyclohexane Distillation

Benzene, toluene, Ethyl benzene, Xylene Catalytic reforming

Alkyl benzenes Akylation

>C9 Polymerisation

2

The diagram above schematically depicts the major hydrocarbon sources and processes used in producing petrochemicals.

Petrochemical feedstock sources

Like commodity chemicals, petrochemicals are made on a very large scale. Petrochemical manufacturing units differ from commodity chemical plants in that they often produce a number of related products. Compare this with specialty chemical and fine chemical manufacture where products are made in discrete batch processes.

The petrochemicals industry is competitive, involves significant technological innovation and is capital intensive. Currently the industry has pervaded the entire spectrum comprising each and every sphere of our life from food to water security; health care to social and physical

infrastructure; household items to construction automobiles to electronics and communication; agriculture to horticulture, etc.

Historically the industry evolved out of technological innovation in the developed industrialised economies. Until the last quarter of the twentieth century, production of petrochemicals was concentrated in Western Europe, the United States and Japan. Over the last few decades, however, production in areas with competitively priced feedstocks has increased dramatically.

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New production capacity has been built in the Middle East and Asia. Nigeria currently has six petrochemical industries producing different feedstocks for other industries.

MECHANISM, SYNTHESIS AND APPLICATIONS OF ORGANIC REACTIONS

Organic reactions involve the transformation of organic molecules through specific mechanisms, often categorized as addition, elimination, substitution, rearrangement, and redox reactions, and are used in synthesis to create new complex molecules with applications in pharmaceuticals, plastics, and other materials; the mechanisms detail the step-by-step movement of electrons during a reaction, allowing chemists to predict and control the products formed.

BENZENE

Benzene is a colourless liquid (b.p. 80oC) with a characteristic aromatic smell. It is immiscible with water, forming the upper of two layers but is soluble in organic solvents as well being a useful solvent itself. Like all aromatic compounds it burns with smoky and luminous flame owing to its relatively high carbon content. Benzene is a highly toxic compound and continued inhalation of the vapours is dangerous and can induce anaemia and even leukemia

SUBSTITUTION REACTIONS OF BENZENE

Mechanism/ Reactions

The availability of π electrons in benzene serves as a source of electrons i.e. a nucleophilic reagent and its reactions are characterized by electrophilic substitution in the ring. In this and other ring substitution reactions, an unstable intermediate is formed in which the entering atom or group and the leaving hydrogen atom are momentarily bonded to the ring. This causes a disruption in the symmetry of delocalized π electrons and may be illustrated by an incomplete circle inside the ring. If the attacking species is an electrophile, which is most often common, then the reaction intermediate will carry a positive charge which is distributed over the bulk of the molecules. Similarly, in the case of nucleophilic substitution, a negative charge is distributed over the molecule.

Z Z

Y H

1.

Electrophilic attack Nucleophilic attack

NITRATION

4

A 'nitrating mixture', containing equal molar quantities of concentrated nitric acid and sulphuric acids carefully mixed together is added to the benzene. The mixture is then refluxed in a water

bath maintained at 55-600C.

The weaker nitric acid functions as a base in the highly acidic sulphuric acid medium releasing the nitryl (nitronium) cation, NO_2^+ , which then undergoes substitution in the ring.



NO_2

H

NO_2

$\text{NO}_2^+ \text{HSO}_4^-$

2. + H_2SO_4

Slow Fast

Nitrobenzene

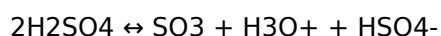
Kinetic studies show that the overall rate determining step is the one involving the attack of the nitryl cation on the ring (Rate $\propto [\text{C}_6\text{H}_6][\text{NO}_2^+]$).

Concentrated nitric acid contains very little of the nitryl cation, and hence it has hardly any effect on benzene in the absence of sulphuric acid.

Nitrobenzene is a pale yellow liquid (b.p. 210oC) with smell of almond, but during its preparation small quantities of white crystals of 1, 3 - dinitrobenzene (m.p. 900C) are almost invariably formed.

SULPHONATION

Sulphonation of benzene to form benzene sulphonic acid is best carried out using fuming sulphuric acid (concentrated sulphuric acid containing excess of sulphur(vi) oxide(sulphur trioxide). The reaction takes place on a gentle warming to 35-500C. Alternatively, ordinary concentrated sulphuric acid could be used but the mixtures requires refluxing for several hours. The most probable mode of attack seems to be electrophilic substitution by the electron deficient sulphur(VI) oxide, SO_3^+ , although the SO_3H^+ ion has been considered as a probability.



$\text{SO}_3 - \text{SO}_3\text{H}$

H

$\text{SO}_3 -$

SO_3

3.

+ H-

Benzene Sulphuric acid

The overall reaction is reversible with the equilibrium tending towards the left.

5

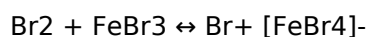
HALOGENATION

Substitution by chlorine or bromine in benzene takes place readily at room temperature in the presence of a 'halogen carrier' catalyst, e.g. Fe, Al, but iodine, I does not easily substitute in this way. The catalyst functions as a Lewis acid, inducing a certain degree of polarity in the halogen molecule.

Although metallic catalysts are commonly used, they are undoubtedly converted into the halide during the reaction and it is in this form they promote the reaction. Consequently, the catalyst may be introduced as the appropriate metal halide.

Bromination may be carried out using an iron or iron (III) bromide catalyst, which polarizes the bromine molecule forming the complex, $\text{Br}^{\delta+} \cdots \text{Br}^{\delta-} \cdots \text{FeBr}_3$ which then ionizes to $\text{Br}^+[\text{FeBr}_4]^-$. It is not possible to state precisely, if at all, to what extent free Br^+ is present.

The reaction may be represented in the following manner:



H Br

Br

4 $\text{Br}^+[\text{FeBr}_4]^-$

$[\text{FeBr}_4]^-$

+ HBr + FeBr_3

Bromobenzene

Under prolonged conditions and in the presence of higher proportions of the halogen, the 1, 4-dihalide is formed.

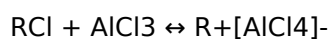
In the presence of direct sunlight or ultraviolet radiation benzene reacts with chlorine or bromine at room temperature to form white crystals of the 1, 2, 3, 4, 5, 6-hexachlorocyclohexane or hexabromocyclohexane respectively



This is an addition reaction and proceeds via free radical mechanism.

FRIEDEL- CRAFT ALKYLATION

In the presence of anhydrous aluminum chloride catalyst, benzene condenses with haloalkane to yield alkyl benzene.



6

R

H

R

$\text{R}^+ [\text{AlCl}_4]^-$

5

$[\text{AlCl}_4]^-$

+ AlCl_3 + HCl

Alkylbenzene

Unfortunately the reaction is very difficult to stop at this stage since alkyl benzene is more reactive than benzene itself and further alkylation of the ring occurs.

Other suitable catalyst include SnF_4 , BF_3 and HF.

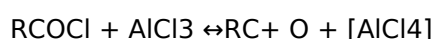
Benzene can also be alkylated with olefins. Alkylation of benzene with olefins refers to a chemical reaction where an alkyl group (derived from an olefin) is added to a benzene molecule, typically using a catalyst like a Lewis acid, resulting in an alkylbenzene product; this process is

considered an electrophilic aromatic substitution reaction as shown in the earlier reaction mechanism above, and is often used to produce important chemicals like linear alkylbenzene (LAB) for detergent production.

The alkylation of benzene with 1-dodecene was accomplished using zirconia supported silicotungstic acid (STA/ZrO₂) as the catalyst as the traditional Aluminum Chloride and hydrofluoric acid catalysts have been found to be hazardous.

FRIEDEL - CRAFT ACYLATION

The condensation of benzene with an acyl chloride, RCOCl or acid anhydride, (RCO)₂O, in the presence of an anhydrous aluminum chloride catalyst gives good yield of the aromatic ketone. The mixture is refluxed on a water bath at 500 C



In the case of the anhydride, (RCO)₂O + AlCl₃ $\rightleftharpoons$ RC⁺ + O⁻ + [AlCl₄]⁻

7

COR

H

COR

RC O

6

[AlCl₄]⁻

+ AlCl₃ + HCl

Aromatic ketone

The reaction with the anhydride is generally preferred because the anhydride is more easily obtained in a high state of purity, is more and easily handled and the resulting ketone is more easily separated.

ALKYLATION OF PHENOLS WITH OLEFINS

Alkylation of phenols with olefins is a chemical reaction where an olefin (alkene) molecule is added to a phenol molecule, resulting in the formation of an alkylphenol, typically catalyzed by an acid catalyst; this process involves the electrophilic attack of the phenol ring by the positively charged carbon species generated from the olefin, leading to the attachment of the alkyl group to the aromatic ring of the phenol.

Alkylation of phenol with 1-dodecene is important as it is used for the production of dodecylphenol, a dense viscous, light yellow liquid with the characteristic smell. Dodecylphenol is used as raw material for manufacturing lubricant additives, anionic detergents, antioxidants, and in phenol resins, adhesives, paints and accelerator for curing epoxy resins.

POLYMERISATION OF ETHYLENE

The polymerisation of ethylene (ethene), involves the joining together of thousands of ethene molecules to form poly (ethylene) - commonly called polythene. The reaction is done at high pressures in the presence of a trace of oxygen as an initiator and iron (Fe) used as catalyst.

This reaction occurs via a free radical mechanism involving some steps as outlined below:

Chain initiation

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The oxygen reacts with some of the ethene to give an organic peroxide. Organic peroxides are very reactive molecules containing oxygen-oxygen single bonds which are quite weak and which break easily to give free radicals. You can short-cut the process by adding other organic peroxides such as benzoyl peroxide, directly to the ethene instead of using oxygen if you want to. The type of the free radicals that start the reaction off vary depending on their source. For simplicity we give them a general formula: $Ra\bullet$

Chain propagation

In an ethene molecule, $CH_2=CH_2$, the two pairs of electrons which make up the double bond aren't the same. One pair is held securely on the line between the two carbon nuclei in a bond called a sigma bond. The other pair is more loosely held in an orbital above and below the plane of the molecule known as a π bond.

The free radical ($Ra\bullet$) approaches the π bond in ethene

The free radical, Ra , uses one of the electrons in the π bond to help to form a new bond between itself and the left hand carbon atom. The other electron returns to the right hand carbon.

This is energetically worth doing because the new bond between the radical and the carbon is stronger than the π bond which is broken. You would get more energy out when the new bond is made than was used to break the old one. The more energy that is given out, the more stable the system becomes. What we've now got is a bigger free radical - lengthened by CH_2CH_2 . That can react with another ethene molecule in the same way:

So now the radical is even bigger. That can react with another ethene - and so on and so on. The polymer chain gets longer and longer.

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Chain termination

The chain does not, however, grow indefinitely. Sooner or later two free radicals will collide together.

That immediately stops the growth of two chains and produces one of the final molecules in the poly(ethylene). It is important to realise that the poly(ethylene) is going to be a mixture of molecules of different sizes, made in this sort of random way.

Chemicals produced from ethylene

- ethylene - the simplest olefin; used as a chemical feedstock and ripening stimulant
- polyethylene - polymerized ethylene; LDPE, HDPE, LLDPE
- ethanol - via ethylene hydration (chemical reaction adding water) of ethylene
- ethylene oxide - via ethylene oxidation
- ethylene glycol - via ethylene oxide hydration
- engine coolant - ethylene glycol, water and inhibitor mixture
- polyesters - any of several polymers with ester linkages in the main chain
- glycol ethers - via glycol condensation

- ethoxylates
- vinyl acetate
- 1,2-dichloroethane
- trichloroethylene
- tetrachloroethylene – also called perchloroethylene; used as a dry cleaning solvent and degreaser
- vinyl chloride – monomer for polyvinyl chloride
- polyvinyl chloride (PVC) – a type of plastic used for piping, tubing, other things

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Chemicals produced from propylene

- propylene – used as a monomer and a chemical feedstock
- isopropyl alcohol – 2-propanol; often used as a solvent or rubbing alcohol
- acrylonitrile – useful as a monomer in forming Orlon, ABS
- polypropylene – polymerized propylene
- polyether polyol – used in the production of polyurethanes
- propylene glycol – used in engine coolant and aircraft deicer fluid
- glycol ethers – from the condensation of glycols
- acrylic acid
- acrylic polymers
- allyl chloride
- epichlorohydrin – chloro-oxirane; used in epoxy resin formation
- epoxy resins – a type of polymerizing glue from bisphenol A, epichlorohydrin, and some amine

Butene

- isomers of butylene – useful as monomers or co-monomers
- isobutylene – feed for making methyl tert-butyl ether (MTBE) or monomer for copolymerization with a low percentage of isoprene to make butyl rubber

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- 1,3-butadiene (or buta-1,3-diene) – a diene often used as a monomer or co-monomer for polymerization to elastomers such as polybutadiene, styrene-butadiene rubber, or a plastic such as acrylonitrile-butadiene-styrene (ABS)
- synthetic rubbers – synthetic elastomers made of any one or more of several petrochemical (usually) monomers such as 1,3-butadiene, styrene, isobutylene, isoprene, chloroprene; elastomeric polymers are often made with a high percentage of conjugated diene monomers such as 1,3-butadiene, isoprene, or chloroprene.

Aromatics

Chemicals produced from benzene

- benzene – the simplest aromatic hydrocarbon
- ethylbenzene – made from benzene and ethylene

- styrene – made by dehydrogenation of ethylbenzene; used as a monomer
- polystyrenes – polymers with styrene as a monomer
- cumene – isopropylbenzene; a feedstock in the cumene process
- phenol – hydroxybenzene; often made by the cumene process
- acetone – dimethyl ketone; also often made by the cumene process
- bisphenol A – a type of "double" phenol used in polymerization in epoxy resins and making a common type of polycarbonate
- epoxy resins – a type of polymerizing glue from bisphenol A, epichlorohydrin, and some amine
- polycarbonate – a plastic polymer made from bisphenol A and phosgene (carbonyl dichloride)
- solvents – liquids used for dissolving materials; examples often made from petrochemicals include ethanol, isopropyl alcohol, acetone, benzene, toluene, xylenes
- cyclohexane – a 6-carbon aliphatic cyclic hydrocarbon sometimes used as a non-polar solvent
- nitrobenzene – can be made by single nitration of benzene
- aniline – aminobenzene
- alkylbenzene – a general type of aromatic hydrocarbon, which can be used as a precursor for a sulfonate surfactant (detergent)
- detergents – often include surfactants types such as alkylbenzene sulfonates and nonylphenol ethoxylates
- chlorobenzene

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Primary petrochemicals are produced directly from petroleum or natural gas and include ethylene, propylene, butadiene, benzene, toluene and xylene. These chemicals are then further processed to produce secondary petrochemicals such as polyethylene, polypropylene, polystyrene and PVC

MANUFACTURE OF SYNTHETIC RUBBER

The manufacture of synthetic rubber begins by refining a hydrocarbon mixture, typically derived from oil or coal, to produce monomers like butadiene, styrene, and isoprene; these monomers are then polymerized using a catalyst to form long polymer chains, creating the rubber substance, which can be further processed through vulcanization to enhance its properties and finally shaped into desired products depending on the application; key steps include refining the source material, creating polymers, vulcanization, and shaping.

Different types of synthetic rubber and their common monomers:

- Styrene-butadiene rubber (SBR): Made from styrene and butadiene
- Nitrile rubber (NBR): Made from butadiene and acrylonitrile
- Butyl rubber (IIR): Made from isoprene and isobutylene
- Neoprene: Made from chloroprene

The most prevalent synthetic rubber is styrene-butadiene rubbers (SBR) derived from the copolymerization of styrene and 1,3-butadiene. Other synthetic rubbers include: polyisoprene,

prepared by polymerization of isoprene; neoprene, prepared by polymerization of 2-chlorobutadiene.

OXIDATION OF HYDROCARBONS

Hydrocarbon oxidation is a chemical reaction that breaks down hydrocarbons into carbon dioxide and water. It's a chain reaction that's initiated by free radicals.

Aerobic oxidation of hydrocarbons is crucial in industrial petrochemical processes. In most cases, reducing agents and/or radical initiators have been used under severe conditions.

The oxidation of hydrocarbons is a free radical-initiated autocatalytic chain reaction. The reaction is slow at the start and accelerates with increasing concentration of the resulting hydroperoxides.

In addition, 'free' carbon-centered radicals react rapidly with so far undamaged polymer segments as well as with oxygen under the formation of peroxide radicals that itself attack further intact polymer chain segments leading to those hydroperoxides and other carbon-centered radicals. The rate constant for the reaction of most alkyl radicals with oxygen is of the order of 10^7 – $10^9 \text{ mol}^{-1} \text{ s}^{-1}$.

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As formulated for general radical-based degradation pathways, this process can be regarded as proceeding in three distinct steps: chain initiation, chain propagation including chain branching, and chain termination

Hydration of Olefins

When alkenes are treated with aqueous acid (H_3O^+) they can be converted to alcohols. This is a hydration process, a chemical reaction that adds water to an olefin (alkene) to produce an alcohol. It's an industrial process that uses an acid catalyst, such as sulfuric or phosphoric acids

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

EFFURUN DELTA STATE, NIGERIA

SYNTHETIC RUBBER

BY

INDUSTRIAL CHEMISTRY STUDENT

(GROUP A)

PRESENTED TO

MR. SAMUEL ASNWANI

MARCH 2025

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ABSTRACT

Historical analysis of synthetic rubber production from divinyl from 1903 to 2014 in Russia, based on ethanol, the relevance of which use as an alternative petroleum source dictated projected exhaustion of oil resources. The necessity of a technology Nano structure catalyst development for a new generation, as well intensification of ethanol conversion process into divinyl by using initiators is demonstrated.

INTRODUCTION

Synthetic rubber is an artificial elastomer synthesized from petroleum-based chemicals, engineered to mimic or surpass the properties of natural rubber (derived from the latex of the Heave Brasiliense's tree). Developed to address limitations in natural rubber supply and performance, it plays a crucial role in modern industry. Its history, production, types, applications, and environmental considerations highlight its significance and ongoing evolution.

HISTORICAL BACKGROUND

The quest for synthetic rubber began in the late 19th century as industrial demand outpaced natural rubber production. Early experiments, like Fritz Hofmann's polymethyl rubber in 1909, laid the groundwork. The real breakthrough came during World War II, when Japan's control of Southeast Asian rubber plantations cut off natural supplies. The United States and allies scaled up production of styrene-butadiene rubber (SBR), marking synthetic rubber's rise. Since then, advancements have tailored it for diverse uses.

PRODUCTION PROCESS

Synthetic rubber starts with petrochemical monomers such as butadiene, styrene, or isoprene. The process involves:

Polymerization: Monomers are chemically bonded into long polymer chains using methods like emulsion (water-based), solution (solvent-based), or suspension polymerization.

Catalysts and controlled conditions ensure the desired structure.

COMPOUNDING: Additives—carbon black for strength, oils for flexibility, and antioxidants for longevity—are blended in.

VULCANIZATION: Sulfur or other agents cross-link the polymer chains, enhancing elasticity and durability.

This energy-intensive process allows customization, unlike the fixed properties of natural rubber.

TYPES OF SYNTHETIC RUBBER

NITRILE RUBBER

Nitrile rubber, also known as nitrile butadiene rubber, NBR, Buna-N, and acrylonitrile butadiene rubber, is a synthetic rubber derived from acrylonitrile and butadiene. Trade names include Perbunan, Nipol, Krynac and Europrene. This rubber is unusual in being resistant to oil, fuel, and other chemicals.

Nitrile rubber was first developed in the 1930s.

It was initially used as a substitute for natural rubber during World War II

PROPERTIES

- Oil and chemical resistance: Nitrile rubber is resistant to many liquids and gases, including petroleum oils, hydrocarbon fuels, and dry-cleaning fluids.
- Tensile strength: Nitrile rubber has high tensile strength.
- Abrasion resistance: Nitrile rubber has good abrasion resistance.
- Puncture resistance: Nitrile gloves are more puncture-resistant than natural rubber gloves.
- Allergic reaction: Nitrile rubber is less likely to cause an allergic reaction than natural rubber.

USES

- Gaskets: Nitrile rubber is used for gaskets.
- Gloves: Nitrile rubber is used for gloves.
- Automotive: Nitrile rubber is used in oil hoses, seals, grommets, and more.
- Aerospace: Nitrile rubber is used in the aerospace industry.
- Hydraulic hoses: Nitrile rubber is ideal for hydraulic hoses

SILICONE RUBBER

Silicone rubber is an elastomer composed of silicone-itself a polymer-containing silicon together with carbon, hydrogen, and oxygen. Silicone rubbers are widely used in industry, and there are multiple formulations.

PROPERTIES

Silicone rubbers are often one- or two-part polymers, and may contain fillers to improve properties or reduce cost. Silicone rubber is generally non-reactive, stable, and resistant to extreme environments and temperatures from -55 to 300 °C (-70 to 570 °F) while still maintaining its useful properties.

- Temperature resistance: Silicone rubber is flexible at low temperatures and resistant to heat at high temperatures.
- Chemical resistance: Silicone rubber is resistant to many chemicals, including solvents, dilute acids, and bases.
- Water resistance: Silicone rubber has a water-repellent surface.
- Flame resistance: Silicone rubber is flame resistant and produces non-toxic combustion products in a fire.
- Radiation resistance: Silicone rubber is resistant to radiation.

Some Other characteristics are

- it is easy to process.
- Silicone rubber has a neutral taste and odor.
- Silicone rubber is durable and can take a long

USES

- **Molds:** Silicone rubber is used to make elastic molds for a variety of applications.
- **Insulation:** Silicone rubber is used to insulate wire and cables.
- **Hoses and belts:** Silicone rubber is used to make hoses and belts.
- **Springs and mounts:** Silicone rubber is used to springs and flexible molds
- **Adhesives:** Silicone rubber is used as an adhesive to bond surfaces together

Polychloroprene (Neoprene): Weather-, ozone-, and flame-resistant; used in wetsuits, gaskets, and cables.

polybutadiene: Highly resilient, enhancing tire treads and sporting goods like golf balls.

Styrene-Butadiene Rubber (SBR): The most widely used, prized for abrasion resistance in tires, footwear, and belts.

Others: EPDM (weatherproof, used in roofing), butyl rubber (airtight, used in inner tubes), and more

ADVANTAGES OF SYNTHETIC RUBBER

Synthetic rubber's edge lies in its versatility. It can be engineered for heat resistance (silicone), chemical stability (nitrile), or durability (SBR), often outperforming natural rubber in harsh conditions. It's not weather-dependent, ensuring a steady supply. However, natural rubber retains superior elasticity and tensile strength in some cases, though it falters against oils and UV exposure.

APPLICATIONS

Synthetic rubber is ubiquitous:

- **Automotive:** Tires (over 60% of synthetic rubber use), belts, hoses, and gaskets.
- **Construction:** Roofing membranes, sealants, and vibration dampeners.
- **Consumer Goods:** Shoe soles, mattresses, and elastic bands.
- **Industrial:** Conveyor belts, coatings, and adhesives.
- **Medical:** Gloves, tubing, and prosthetics.

ENVIRONMENTAL IMPACT

Synthetic rubber's reliance on fossil fuels raises concerns. Production generates CO₂ emissions, and vulcanized rubber resists biodegradation and recycling. Progress includes:

Bio-based Alternatives: Rubbers from biomass or sugarcane reduce oil dependency.

Recycling: Devulcanization breaks cross-links, enabling reuse.

Greener Processes: Lower-energy methods and carbon capture are emerging.

FUTURE PROSPECTS

Innovations promise smarter synthetic rubbers—self-healing polymers, nanocomposites for strength, and sustainable feedstocks. The challenge is balancing performance, cost, and eco-friendliness. As demand grows, especially in automotive and industrial sectors, synthetic

rubber will evolve toward a circular economy.

CONCLUSION

Synthetic rubber is a cornerstone of modern life, born from necessity and refined through ingenuity. Its adaptability meets diverse needs but substitution ability remains a work in progress from tires to medical tools. It's a material that keeps us moving literally and figuratively.

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